ENKU Base R144 | NO FAB

Root project

ENKU R144 ELECTRICAL ARCHITECTURE

Base R144 engineering source — NO FAB. Physical qualification is pending. No dock/frontlight/Hall/Qi.

Power



File: power.kicad_sch

MCU + IO



File: mcu_io.kicad_sch

EPD + HV



File: epd_hv.kicad_sch

Connectors

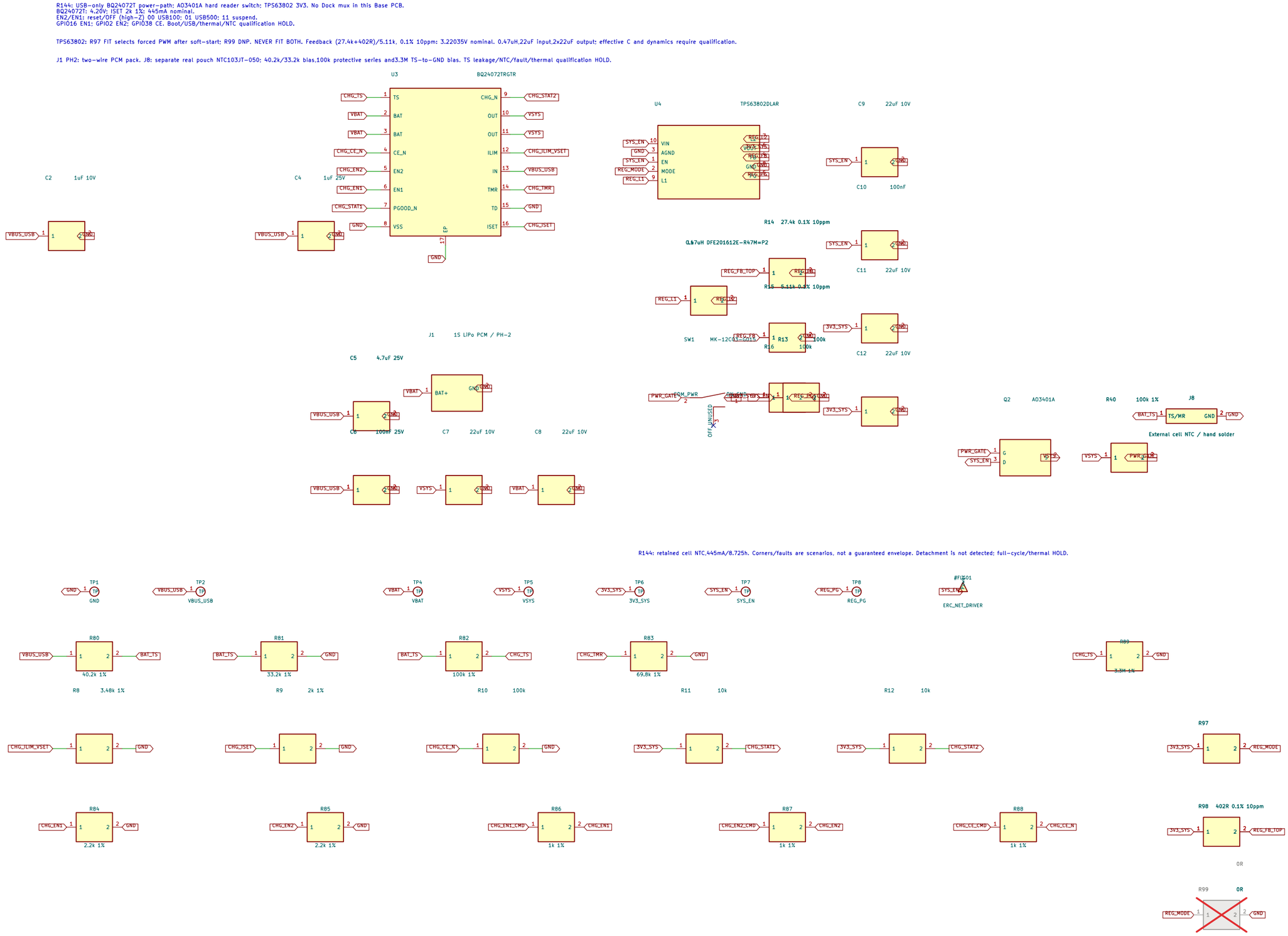


File: connectors.kicad_sch

Five-sheet actual R144 hierarchy. Physical tests NOT RUN.

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Power



Source 676c157fb58c1d5b689b... | native KiCad 10.0.7 | 2/5

Five-sheet actual R144 hierarchy. Physical tests NOT RUN.

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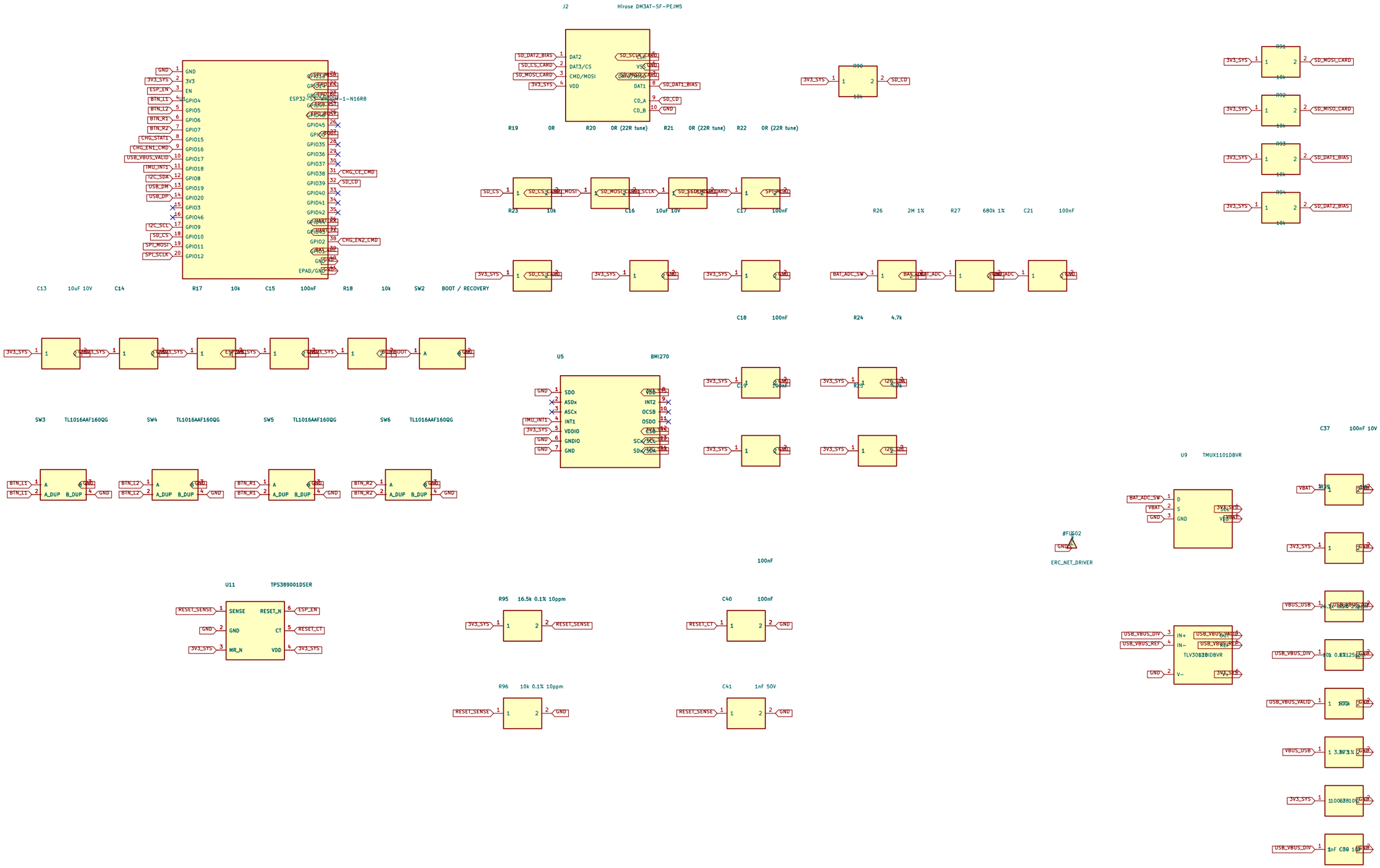
MCU and IO

BMI270 is wired in I2C mode: SDO=GND (default address), CSB=3V3, VDD/VDDIO=3V3, INT1 only; unused secondary/OIS pins are NC.

BASE R144: Dedicated PCB — Hall sensor and cover circuitry removed (not DNP); no wireless charging, no frontlight. Separate Pro PCB.

microSD stays on shared SPI with independent CS and 0R/22R tuning links. All CMD/DAT0..3 have external10k; DAT1/DAT2 are bias-only in SPI.

Battery ADC divider 2M/680k draws ~1.6uA at 4.2V; 100nF at ADC node provides acquisition charge reservoir. Validate ADC settling/calibration in firmware.



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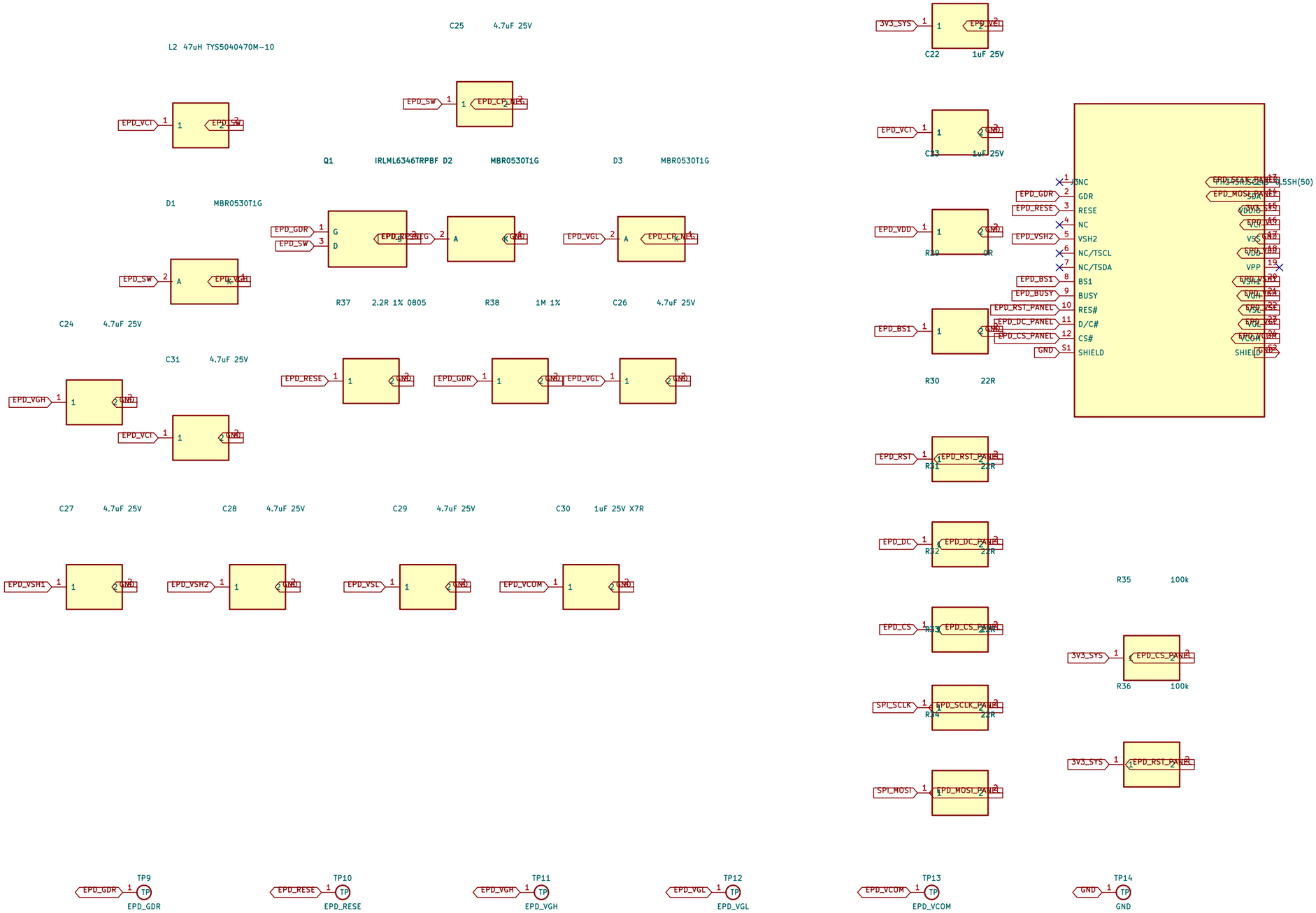
E-paper HV

SSD1677 application circuit captured as discretes. R144 retains reference from GDEY0397T81P supplied datasheet p19: 47uH + NMOS + 3x MBR0530 + 2.2R RESE + 1M GDR pulldown + local 4.7uF input capacitor.

D1: SW -> VGH. C25 flying capacitor + D2/D3 form the negative VGL pump. Q1 source is sensed at RESE through 2.2R to GND.

VPP is explicitly left NC for normal operation; BS1 is tied LOW for 4-wire SPI. Current panel pins 6/7 remain NC pending panel-specific continuity confirmation.

R144 requires first-board scope validation of VGH/VGL/VSH/VSL/VCOM. 47uH is the documented panel reference; 10uH is not an approved default. Pin5 VDHR=VSH2 confirmed by Felix10Oct2026; FPC fit still HOLD.



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Five-sheet actual R144 hierarchy. Physical tests NOT RUN.

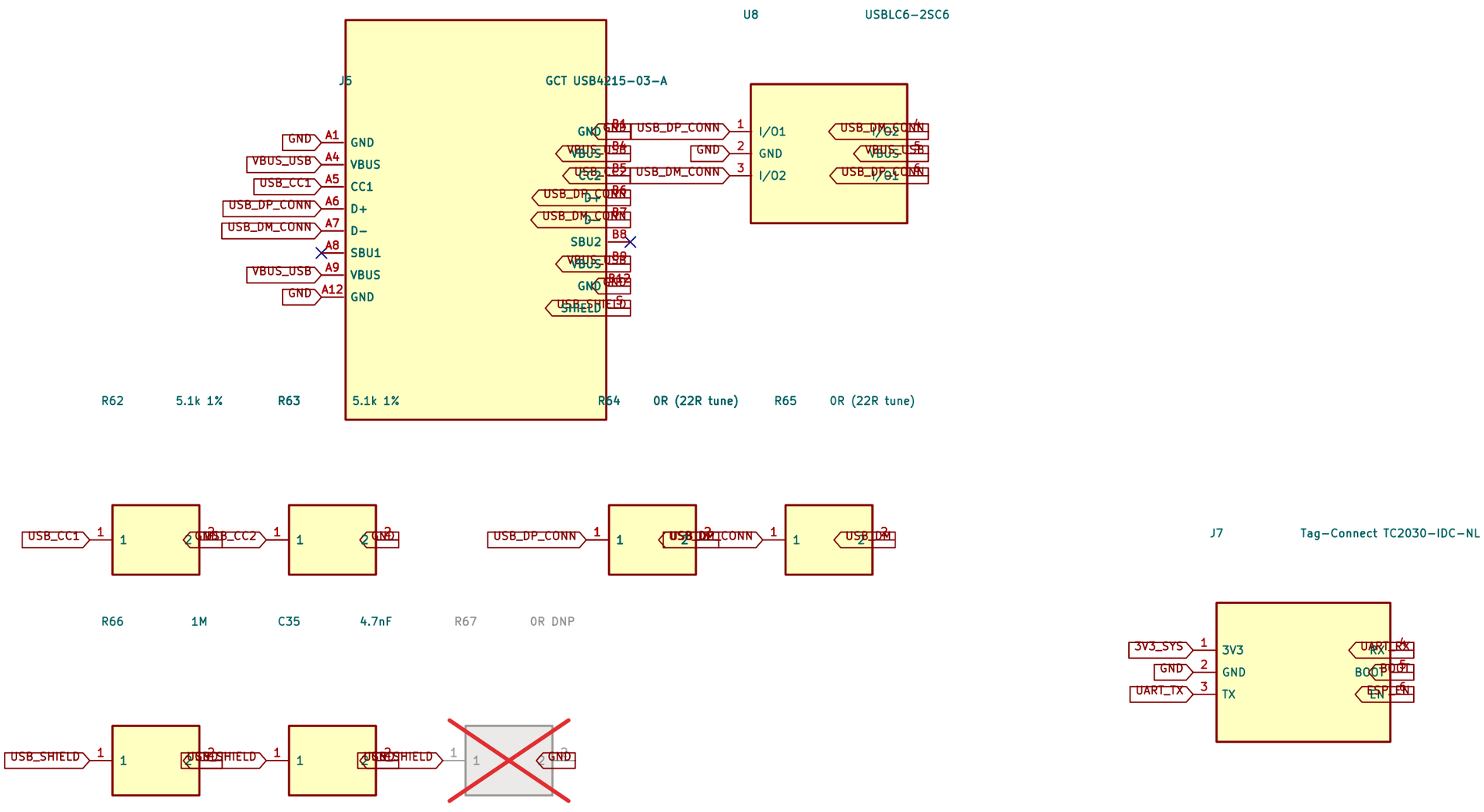
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Connectors

USB-C is now captured as the full 16-pin USB2 receptacle: all VBUS/GND pins joined, 5.1k Rd on CC1/CC2, SBU left NC.

D+/D- get an ESD footprint and populated 0R tuning links; the links can be changed to 22R after SI/EMI validation without a respin.

Dock detect retains its 10k series / 100k pulldown / 10nF filter baseline. Qi is removed from Base R0.1 and deferred to a later Pro/Wireless revision.



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Five-sheet actual R144 hierarchy. Physical tests NOT RUN.